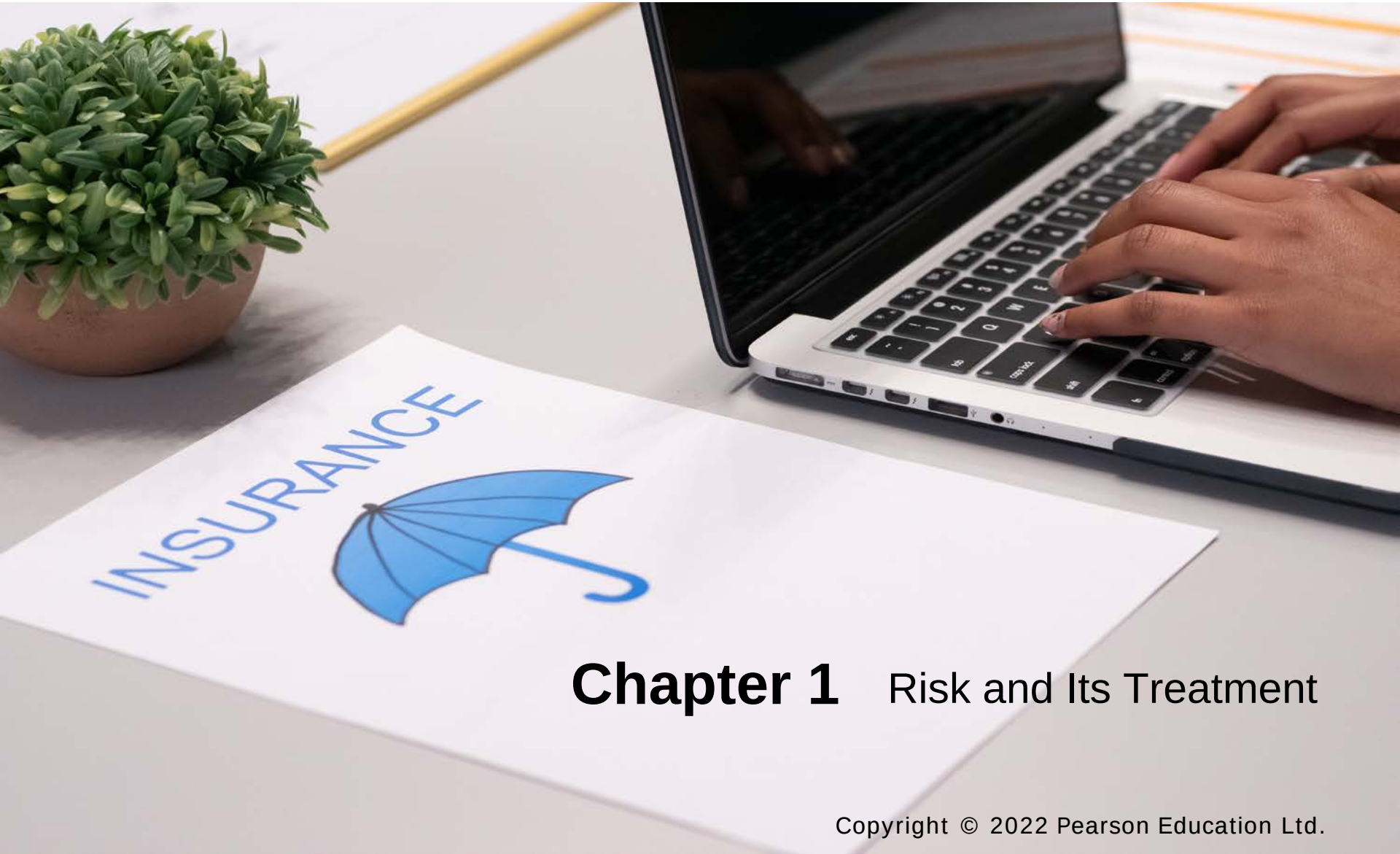


Principles of Risk Management and Insurance



Chapter 1 Risk and Its Treatment

Learning Objectives (1 of 2)

1.1 Explain the historical definition of risk(風險/危險).

1.2 Explain the meaning of loss exposure(損失暴露/曝險).

1.3 Understand the following types of risk:

- Pure risk 純粹風險
- Speculative risk 投機風險
- Diversifiable risk 可分散風險
- Nondiversifiable risk 不可分散的風險
- Enterprise risk 企業風險
- Systemic risk 系統性風險

Learning Objectives (2 of 2)

1.4 Identify the major pure risks that are associated with great economic insecurity.

1.5 Show how risk is a burden to society.

1.6 Explain the major techniques for managing risk.

What is Risk? 甚麼是風險
What are the risks in our life?
現實生活中有哪些風險

地震 颱風 水災
暴雪 龍捲風
車禍 火災



Definitions of Risk (1 of 2)

- Traditional Definition of **Risk**: Uncertainty concerning the occurrence of a loss
 - 損失發生的不確定性: 1. 不確定會不會發生 2. 不確定何時發生 3. 不確定發生後的損失
- In the insurance industry, **risk** is also used to identify the property or life that is being considered for insurance
 - 保險產業所謂的風險，必須符合大數法則 (law of large numbers)，可以定價，才可能成為可保風險
- In economics and finance:
 - The term **risk** is used in situations where the probabilities of possible outcomes are known
 - **Uncertainty** is used in situations where such probabilities cannot be estimated

Definitions of Risk (2 of 2)

- **Loss Exposure(損失暴露/曝險):** Any situation or circumstance in which a loss is possible, regardless of whether a loss occurs (任何發生可能損失的狀況但不管此狀況有無發生損失)
- **Objective risk (客觀風險)** is defined as the relative variation of actual loss from expected loss (實際損失與預期損失的相對差異)
 - It can be statistically calculated by some measure of dispersion, such as the standard deviation
- **Subjective (perceived) risk (主觀風險)** is defined as uncertainty based on a person's mental condition or state of mind

Chance of Loss (1 of 2)

- **Chance of loss:** The probability that an event that causes a loss will occur (損失機率)
- **Objective probability** refers to the long-run relative frequency of an event based on the assumptions of an infinite number of observations and of no change in the underlying conditions (客觀機率，長期事件發生的期望值)
- **Subjective probability** is the individual's personal estimate of the chance of loss (主觀機率)

Chance of Loss (2 of 2)

- Chance of loss is the probability that an event that causes a loss will occur.
- Objective risk is the relative variation of actual loss from expected loss

The chance of loss may be identical for two different groups, but objective risk may be quite different!

City	# homes	Chance of Fire	Average # fires	Range	Objective Risk
Philadelphia	10,000	1%	100	75 – 125	25%
Los Angeles	10,000	1%	100	90 – 110	10%

Peril and Hazard (1 of 2)

- A **peril** (險境/險源/危險事故) is defined as the cause of the loss.
 - 損失發生的原因
 - Examples include property damage because of fire, windstorm, lightning, or damage to your car because of a **collision** with another vehicle.
- A **hazard** (危險要素) is a condition that creates or increases the frequency or severity of loss
 - 增加損失發生頻率或嚴重程度
 - A **physical hazard** is a physical condition that increases the frequency or severity of loss
 - **Moral hazard** is dishonesty or character defects in an individual that increase the frequency or severity of loss

Peril and Hazard (2 of 2)

- **Attitudinal Hazard (Morale Hazard)** is carelessness or indifference to a loss, which increases the frequency or severity of a loss
- **Legal Hazard** refers to characteristics of the legal system or regulatory environment that increase the frequency or severity of loss

Classification of Risk (1 of 5)

- Pure Risk and Speculative Risk
 - A **pure risk** (純粹風險) is a situation in which there are only the possibilities of loss or no loss (earthquake)
 - 純風險是指僅有損失機會而無獲利可能之危險，如火災、地震等
 - A **speculative risk** (投機風險) is a situation in which either profit or loss is possible (gambling)
 - 投機風險是指損失可能，亦有獲利可能，如股票、期貨投資、甚至賭博等

Classification of Risk (2 of 5)

- Diversifiable Risk and Nondiversifiable Risk
 - A **diversifiable risk** (可分散風險) affects only individuals or small groups (car theft). It can be reduced or eliminated by diversification (大量分散).
 - A **nondiversifiable risk** (不可分散風險) affects the entire economy or large numbers of persons or groups within the economy (hurricane). It is also called fundamental risk.
 - Government assistance may be necessary to insure nondiversifiable risks.

Classification of Risk (3 of 5)

- **Enterprise risk (企業風險)** is a term that encompasses all major risks faced by a business firm, which include: pure risk, speculative risk, strategic risk, operational risk, and financial risk
 - **Strategic Risk** refers to uncertainty regarding the firm's financial goals and objectives.
 - **Operational risk** results from the firm's business operations.
 - **Financial Risk** refers to the uncertainty of loss because of adverse changes in commodity prices, interest rates, foreign exchange rates, and the value of money.

參考 Classification of Risk (4 of 5)

- **Enterprise Risk Management** combines into a single unified treatment program all major risks faced by the firm
 - By packaging major risks into a single program, the firm can offset one risk against another.
 - As long as all risks are not perfectly correlated, the firm can offset one risk against another, thus reducing the firm's overall risk.
 - Treatment of financial risks requires the use of complex hedging (避險) techniques, financial derivatives (衍生性金融商品), futures contracts (期貨) and other financial instruments.

Classification of Risk (5 of 5)

- **Systemic risk** is the risk of collapse of an entire system or entire market due to the failure of a single entity or group of entities that can result in the breakdown of the entire financial system

Major Personal 個人風險 and Commercial Risks 企業風險 (1 of 5)

- **Personal risks** are risks that directly affect an individual or family (個人風險是指個人或家庭所面對的風險).
- They involve the possibility of a loss or reduction in income, extra expenses or depletion of financial assets, due to:
 - Premature death 早死風險，家中主要經濟來源者早死或失能，失去收入
 - Retirement risks 退休風險
 - Poor health 健康風險
 - Unemployment 失業
 - Alcohol and drug addiction 酒精或藥物上癮

Major Personal and Commercial Risks (2 of 5)

- **Property risks** (財產風險) involve the possibility of losses associated with the destruction or theft of property 財產風險包含財產破壞或盜竊相關的損失之可能性.
- Direct loss vs indirect loss
 - A **direct loss** is a financial loss that results from the physical damage, destruction, or theft of the property, such as fire damage to a home (直接損失是財產的物理損壞、毀壞或被盜造成的經濟損失，例如房屋火災、颱風、洪水).
 - An **indirect or consequential loss** is a financial loss that results indirectly from the occurrence of a direct physical damage or theft, such as the additional living expenses after a fire (間接或後果性損失是由於發生直接的物理損壞或盜竊而間接造成的財務損失，例如火災後的額外生活費用、火災後的營業中斷費用).

Major Personal and Commercial Risks (3 of 5)

- **Liability risks** (責任風險) involve the possibility of being held legally liable for bodily injury or property damage to someone else (責任風險涉及對於他人的人身傷害或財產損失承擔法律責任的可能性，例如：食品公司需要負擔食品安全責任、遊樂園要負擔遊客安全、旅行社要負擔履約保證、醫師負責醫療疏失責任).
- There is no maximum upper limit with respect to the amount of the loss (損失可能沒上限或非常鉅額)
 - A lien can be placed on your income and financial assets (對於所得或金融資產有扣押權利)
 - Legal defense costs can be enormous (法律訴訟成本非常巨大)

Major Personal and Commercial Risks (4 of 5)

- Commercial Risks:
- Firms face a variety of pure risks that can have serious financial consequences if a loss occurs:
 - **Property risks**, such as damage to buildings, furniture and office equipment
 - **Liability risks**, such as suits for defective products, pollution, and sexual harassment
 - **Loss of business income**, when the firm must shut down for some time after a physical damage loss (營業收入損失，可投保營業中斷保險).
 - **Cybersecurity and identity theft** (網路資安風險) by thieves breaking into a firm's computer system (財產保險公司有推出網路資訊安全保險或是員工竊盜保險)

Major Personal and Commercial Risks (5 of 5)

- Other risks faced by business firms include:
 - **Human resources exposures**, such as job-related injuries (可透過團體保險降低風險)
 - **Foreign loss exposures**, such as acts of terrorism
 - **Intangible property exposures**, such as damage to the market reputation and public image of the company (較無法透過保險分散)
 - **Government exposures**, such as violation of safety standards(較無法透過保險分散)

Burden of Risk on Society

- The presence of risk results in three major burdens on society:
 - In the absence of insurance, individuals and business firms would have to maintain large emergency funds to pay for unexpected losses
 - The risk of a liability lawsuit may discourage innovation, depriving society of certain goods and services
 - Risk causes worry and fear

Techniques for Managing Risk (1 of 4)

- **Risk Control** refers to techniques that reduce the frequency or severity of losses:
 - **Avoidance**
 - **Loss prevention** refers to activities to reduce the frequency of losses
 - **Loss reduction** refers to activities to reduce the severity of losses:
 - **Duplication** (備份)
 - **Separation** (分開)
 - **Diversification** (多樣化)

Techniques for Managing Risk (2 of 4)

- **Risk Financing** refers to techniques that provide for the funding of losses:
 - **Retention** (自留) means that an individual or business firm retains part or all of the losses that can result from a given risk. (個人或企業將損失留給自己承擔)
 - **Active retention** (主動自留) means that an individual is aware of the risk and deliberately plans to retain all or part of it (意味著個人意識到風險並有意計劃保留全部或部分風險)
 - **Passive retention** (被動自留) means risks may be unknowingly retained because of ignorance, indifference, or laziness (意味著由於無知、冷漠或懶惰，風險可能在不知不覺中被保留，無法或沒選擇只能自行承擔)

Techniques for Managing Risk (3 of 4)

- **Self Insurance** is a special form of planned retention by which part or all of a given loss exposure is retained by the firm (自我保險是一種特殊的計劃自留形式，通過這種形式，公司可以保留部分或全部給定的損失風險，又可稱之為self-funding)
- **A Noninsurance transfer** transfers a risk to another party.
 - A transfer of risk by contract, such as through a **hold-harmless clause** in a contract (通過契約轉移風險，例如通過契約中的免責條款).
 - **Hedging** (避險) is a technique for transferring the risk of unfavorable price fluctuations to a speculator (避險是一種將不利價格波動的風險轉移給投機者的技術).
 - **Incorporation** (公司化) of a business firm transfers to the creditors the risk of having insufficient assets (公司化公司將資產不足的風險轉移給股票持有人，企業經營者為有限債權).

Techniques for Managing Risk (4 of 4)

- For most people, **insurance** is the most practical method for handling major risks:
 - Risk transfer is used because a pure risk is transferred to the insurer.
 - The **pooling technique** is used to spread the losses of the few over the entire group
 - The risk may be reduced by application of the **law of large numbers** (當樣本數量愈多，其平均數就會愈趨近於期望值 i.e. 在重複試驗中隨著試驗次數的增加，事件發生的頻率/機率會趨近於一個穩定值)